

Installation Guidelines

This document provides detailed installation guidelines for the following Advance Analytik systems and accessories:

1. Viz Eco Duo – Dual Channel Transmitter
2. Viz Eco Multi – Multi Channel Transmitter
3. VizSens pH (Analog) Sensor
4. VizSens ODO (Optical Dissolved Oxygen) Sensor
5. VizSens SS (Suspended Solids/Sludge) Sensor
6. FRP Mounting Panel
7. SS Flow-through Chamber (for Viz-Eco-EQMS only)

1. Viz Eco Duo – Dual Channel Transmitter

1. Choose a flat surface such as FRP panel or wall for mounting.
2. Use the mounting lugs and screws provided to fix the transmitter securely.
3. Ensure proper sealing (IP65/NEMA4X) when mounted outdoors.
4. Connect power supply: 85-265 VAC or 9-36 VDC.
5. Connect sensor cables to designated input ports (max. 2 sensors).
6. Use 4-20mA and/or MODBUS RS485 output ports for communication with SCADA/PLC.
7. Insert memory card (optional) for data logging if required.

2. Viz Eco Multi – Multi Channel Transmitter

8. Mount on FRP panel or solid wall using mounting lugs provided.
9. Ensure 7" touchscreen is visible and accessible for operators.
10. Connect power: 90-260 VAC or 9-36 VDC.
11. Attach up to 7 sensors using digital/analog ports with corresponding modules.
12. Connect output wiring for 4-20 mA signals, MODBUS RS485, or optional wireless module.
13. Set user access levels as required via on-screen configuration.
14. Check grounding to minimize electrical interference.

3. VizSens pH (Analog) Sensor

15. Mount sensor in tank, channel, or flow-through assembly using threaded or holder design.
16. Ensure immersion depth is sufficient to keep sensing tip submerged.
17. Avoid installation near air bubbles or in turbulent areas.
18. Connect analog cable to Viz Eco transmitter input (ensure polarity and shielding).
19. If using optional converter module, ensure proper digital interface connection.
20. Calibrate using standard pH buffer solutions before operation.

4. VizSens ODO (Optical DO) Sensor

- 21. Mount via immersion in basin or channel, or use supplied holder if applicable.
- 22. Avoid exposure to strong light or air bubble zones.
- 23. Connect RS485 MODBUS output to Viz transmitter securely.
- 24. Ensure membrane cap is intact; replace if worn or expired.
- 25. Allow stabilization time after power-on before reading values.
- 26. Factory calibrated; field calibration optional via transmitter menu.

5. VizSens SS (Suspended Solids/Sludge) Sensor

- 27. Install sensor vertically or at a stable fixed depth in tanks or pipes.
- 28. Use optional automatic cleaning wiper if applicable.
- 29. Ensure consistent flow at 2.5 m/s (if in-line).
- 30. Connect sensor to Viz transmitter via RS485 cable.
- 31. Verify self-diagnosis reports from transmitter interface.
- 32. Avoid areas with heavy turbulence or sedimentation layers.

6. FRP Mounting Panel

- 33. Fix the FRP panel securely to a wall or frame using M8 anchor bolts or clamps.
- 34. Mount transmitters (Viz Eco Duo and Multi) using screws and alignment holes.
- 35. Ensure cable trays or conduits are installed neatly for sensor and power wiring.
- 36. Label sensor channels and power lines for ease of maintenance.
- 37. If installed outdoors, ensure proper shading or enclosure as required.

7. SS Flow-through Chamber (for Viz-Eco-EQMS)

- 38. Mount the chamber on a flat vertical surface using SS brackets or support frame.
- 39. Connect inlet and outlet pipes with compatible unions or threaded fittings.
- 40. Ensure flow rate and pressure are within sensor specifications (≤ 0.1 MPa).
- 41. Insert sensor (e.g., UV, SS, or other inline type) into the chamber port using sealing collar.
- 42. Ensure no air bubbles or bypass leaks affect measurement area.
- 43. Install drain and sample ports for periodic cleaning and flushing.